

Pruthvi Hosamani

Software Engineer | Python & LLM Systems | Backend + ML

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SUMMARY

Python-focused software engineer with hands-on experience across LLM evaluation, RAG systems, and production backend development. Currently building **Arogya AI** — a multilingual medical LLM platform evaluating Groq Llama 3 and MedGemma against Indian clinical guidelines — and leading a full-stack migration of a legacy ASP.NET system to React + .NET 8 for a live client (**RGUHS**). Comfortable navigating large, unfamiliar codebases, Dockerizing dev environments, and triaging real-world bugs. CS Department Topper (CGPA 8.8/10), Drona Prashasti Award recipient.

CORE SKILLS

Languages:	Python (primary), JavaScript/TypeScript, Java, SQL, C
LLM / ML:	LangChain, LangGraph, ChromaDB (Vector DB), RAG architecture, Groq (Llama 3, Whisper), MedGemma, Hugging Face, LLM evaluation & benchmarking, TensorFlow, Keras, PyTorch, YOLOv4
Backend & APIs:	FastAPI, Django, Django REST Framework, Flask, REST API design, JWT auth, OpenAPI, pytest, microservices
Dev Env & Tooling:	Git, GitHub, Docker (Azure SQL Edge on Apple Silicon, containerized local dev), Linux, CI/CD (GitHub Actions), HPC/MapReduce, Postman
Databases:	PostgreSQL, MySQL, MongoDB, SQL Server, ChromaDB, indexing & query optimization
Frontend:	React 19, Angular 18, TypeScript, MUI v7, Tailwind, RxJS

EXPERIENCE

Founder & ML Engineer | Arogya AI — Multilingual Medical LLM Platform

2025 – Present

Self-directed project. LLM evaluation, RAG, multilingual pipelines.

- Built end-to-end **RAG evaluation pipeline** benchmarking Groq Llama 3 against MedGemma on 8 Indian clinical guideline documents; measured retrieval accuracy, factual grounding, and hallucination rate across English, Hindi, and Kannada query sets.
- Architected **LangChain + ChromaDB** vector store with chunking strategies tuned for medical terminology; designed modular **LangGraph router** for intelligent query routing across specialized medical modules (Symptom Checker, Drug Safety, Lab Interpreter).
- Integrated **OpenFDA API** for real-time drug interaction analysis with structured contraindication detection and adverse event lookup.
- Engineered **multilingual voice pipeline** with Groq Whisper STT and edge-tts TTS (Indian voices); deployed FastAPI backend with React + Tailwind frontend.
- Accepted into multiple accelerator application tracks (Sarvam AI, NVIDIA Inception, Microsoft for Startups); code: github.com/pruthvihosmani.

Freelance Software Engineer | RGUHS — Legacy System Migration (Contract)

2026 – Present

Rajiv Gandhi University of Health Sciences. Full-stack migration of a 28-tab affiliation portal.

- Leading end-to-end migration of a legacy **ASP.NET Web Forms** affiliation portal to a modern **React 19 + MUI v7 + Vite + TypeScript** frontend against a **.NET 8** backend — navigating a large, unfamiliar codebase and reverse-engineering 28 wizard tabs with strict field-level parity to legacy HTML as ground truth.
- Set up full local dev environment including **Dockerized Azure SQL Edge** on Apple Silicon (M-series), sqlcmd via Homebrew, and dual-port frontend/backend orchestration (5173/5174 UI, 5000 API).
- Built shared component library (**FormField**, **DataGridTable**, **FileUploadZone**, **UiOnlyTab**) driven by declarative section manifests; wired Redux + redux-persist for form state across tabs and implemented **TabErrorBoundary** for graceful failure isolation.
- Triaged and fixed production bugs in inherited codebase: **SQL column truncation** in AuditLogs (schema migration), **missing slice exports** crashing Tab14/17, and bad MUI default imports crashing Tab18 — all diagnosed by reading stack traces and stepping through unfamiliar legacy logic.
- Generated a 904-line field inventory from real legacy HTML as the single source of truth, enforcing the rule: *never assume fields not evidenced in source* — similar discipline to validating LLM-generated code against ground-truth test harnesses.

- Architected and deployed **Flask-based RESTful APIs** serving 10,000+ industrial records/day with JWT authentication and role-based access control.
- Optimized **PostgreSQL** performance through strategic indexing and query restructuring — reduced average response time from 2.3s to 0.8s (**65% improvement**) across 100,000+ records.
- Implemented anomaly detection (logistic regression + session management), reducing unauthorized access attempts by 85%.
- Delivered 3 production features in Agile sprints with **95%+ test coverage** using pytest; zero critical bugs post-deployment.

ML Engineer Intern | Wolfpack Intelligence LLP

Dec 2022 – Feb 2023

- Developed **YOLOv4** object detection model (TensorFlow/Keras) achieving **95% accuracy**; applied SGD with momentum and Bayesian hyperparameter optimization, improving precision by 10%.
- Built production-grade **Django REST** application for real-time image inference handling 500+ concurrent requests with 99.2% uptime (Gunicorn + Nginx); batch inference pipeline reduced latency by 35%.
- Designed **MongoDB** schema with efficient indexing; defined API contracts via OpenAPI and implemented CORS for React dashboard serving 50+ users.

Research Intern | RV College of Engineering

Sep 2021 – Oct 2021

- Developed SNP-specific primers for SARS-CoV-2 Delta Plus variant using HMMs + PCA; achieved 20% reduction in false positives.
- Processed 20GB+ genomic datasets using **MapReduce** on HPC clusters, improving computational throughput by 50%.

SELECTED PROJECTS

Full-Stack Conduit (Medium.com Clone) | Angular 18, Django REST, PostgreSQL

Mar – May 2023

- Built 15+ Angular components with **RxJS** reactive state management; <100ms page load via lazy loading across 8 modules.
- Designed 12 REST endpoints (Django ViewSets); optimized queries with **select_related/prefetch_related**, reducing N+1 queries by 60%.
- Configured **GitHub Actions CI/CD** for Jasmine/Karma + Django tests, achieving 95%+ coverage across 200+ commits.

Drone-Based Intelligent Magnetic Sensing System | Python, Flask, SVM, LSTM, Kalman Filters

Apr – May 2024

- Designed magnetic anomaly detection using SVM + Gaussian Processes; fused RTK/GPS with Kalman Filters and LSTMs — 30% accuracy gain.
- Built real-time Plotly dashboard with K-Means clustering for high-risk zone segmentation.

Telegram Job Application Bot | Python, Selenium, SQLite, Telegram API

2025

- Multi-file Python automation that listens for job URLs, scrapes Greenhouse/Lever/ATS career pages, and pre-fills Chrome forms via Selenium; tracks applications in SQLite.

EDUCATION

B.E. in Computer Science & Engineering | CGPA: 8.8/10 (Department Topper)

2020 – 2024

Government Engineering College, Koppal, Karnataka

AWARDS & LEADERSHIP

- **Drona Prashasti Award** — Karnataka Government recognition (INR 1,00,000 cash prize) for academic excellence; CSE Department Topper across all 8 semesters.
- Represented VTU at State Level NSS Youth Festival, Raichur; NSS Coordinator (VTU), Red Cross Wing. Certified **Hindi proficiency** (Dakshina Bharat Hindi Prachar Sabha, Parichaya through Rashtrabhasha); native Kannada; professional English.